

Carlos Bautista

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ECE 698C

Project Proposal

The proposed project is a real-time multi-sensor embedded monitoring system that combines an STM32H753ZI microcontroller with a Zybo Z7-20 FPGA. The purpose of the system is to acquire environmental, distance, and motion data using multiple communication protocols, process the information on the microcontroller, and signal system events to the FPGA for hardware monitoring and alert state management. The project will demonstrate hardware/software co-design by dividing the system responsibilities between an embedded system and programmable logic.

The microcontroller will serve as the primary controller and will interface with the following sensors: VL53L1X is a time-of-flight sensor that will provide the distance measurements through I2C, LSM6DS3 is an inertial measurement unit that provides information through SPI communication, SHT41 and BME280 are the environmental sensors whose temperature and humidity are averaged, and four SSD1306 OLED displays will display sensor readings and system status.

The STM32 will communicate with the Zybo Z7-20 through UART at 115200 baud using a custom command protocol. Commands will light up the LEDs on the Zybo to showcase distance alerts, motion events, system-clear operations, and periodic heartbeat messages. The STM32 will use HAL-based peripheral drivers to implement the communication protocols, and the FPGA will implement the programmable logic in VHDL to maintain the current alert state, latch alert history, monitor heartbeat communication for faults, and transmit status information

back to the microcontroller. Testing will include individual sensor validation, I2C multiplexer operation for overlapping addresses for the OLEDs and channel isolation, SPI register communication, UART protocol testing, and threshold-event testing. UART debug logs will be used to assist with the hardware/software integration and general debugging.

The final deliverables will include a working prototype, STM32 firmware, FPGA VHDL source code, system block diagrams, UART protocol documentation, test results, and a final technical report that encompasses this project. The completed project will demonstrate embedded firmware development, digital logic design, multi-protocol communication, sensor integration, fault detection, and STM32-to-FPGA co-design.

